

TrES-3b

R-2769

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220829 · created 2026-09-03T22:59:11+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459820.7959 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.05
Transit depth (Rp/Rs)^2	3.74 +/- 1.94 [%]
Orbital Inclination (inc)	80.73 +/- 2.35
Airmass coefficient 1 (a1)	0.968 +/- 0.014
Airmass coefficient 2 (a2)	0.0152 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	1.45 %
Best Comparison Star	#3 - [337, 429]
Optimal Method	PSF photometry
Transit Duration (day)	0.041 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.193 $\pm$ 0.05 vs 0.16309 (deviation 0.6 σ)
Duration fit vs published	0.041 d vs 0.0567 d (deviation 0.58 σ)
Mid-time O–C	-4.9 min
Depth SNR	3.9
Residual scatter	1.45 %

Light curve

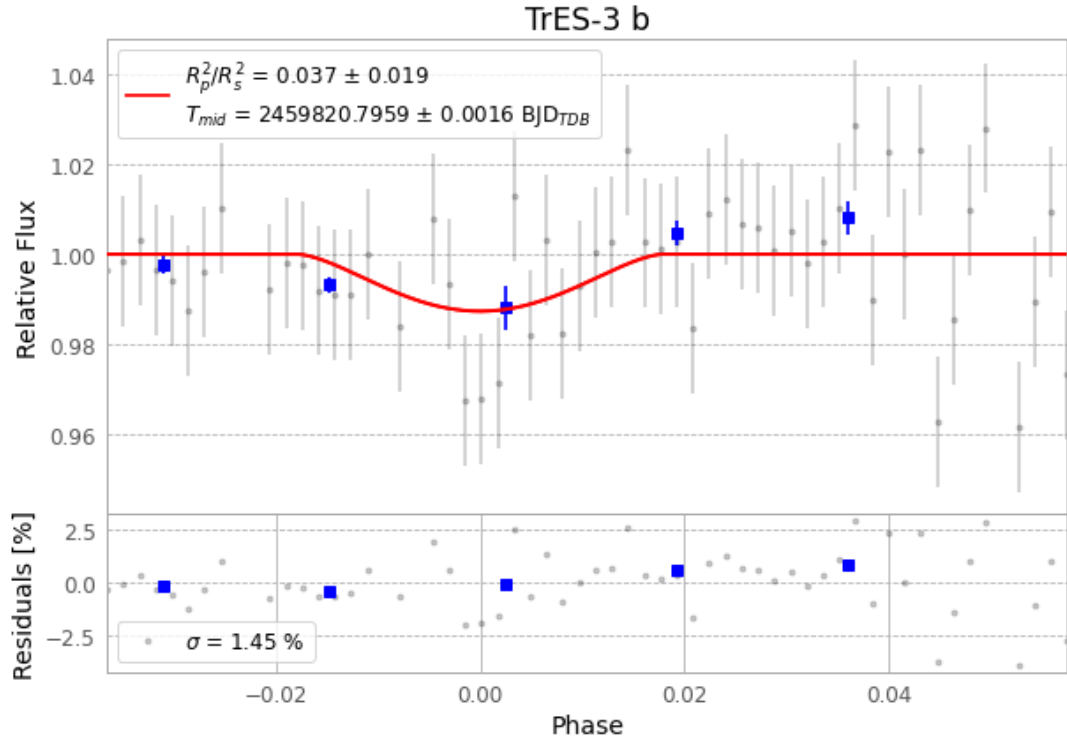


Figure 1 — FinalLightCurve_TrES-3 b_2022-08-28.png (SHA-256 d21e7ddb1e7277f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 216], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 29ea274bfdb3ef77...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

56 FITS frames (37 MB), TRES-3220829055412.FITS → TRES-3220829085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220829.log (a9e5759d7a88...), FinalLightCurve_TrES-3 b_2022-08-28.png (d21e7ddb1e7...), FinalLightCurve_TrES-3 b_2022-08-28.csv (264dc07604c3...)

Compute resources

Wall time 111.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 22:59Z.